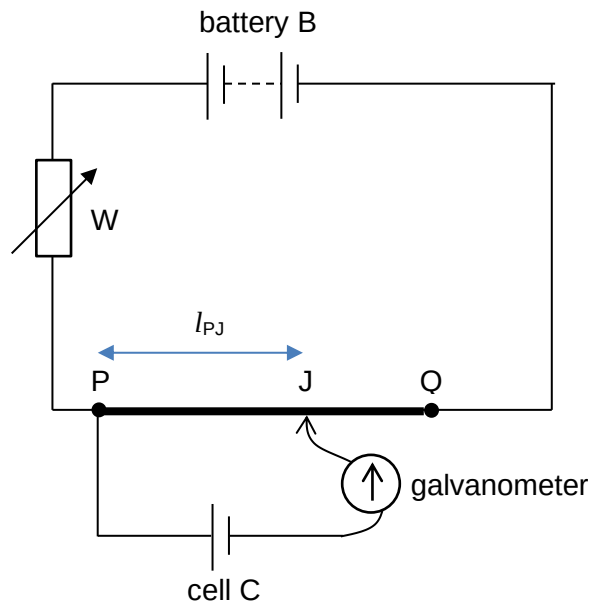


- 23** A battery B, a variable resistor W and a uniform resistance wire PQ are connected in series. A cell C and a galvanometer are connected to the wire PQ with a contact J as shown.



The contact J is moved along wire PQ until the galvanometer reads zero. The distance of J from P, l_{PJ} is then measured.

Which of the following changes will increase the measured distance l_{PJ} ?

- | | |
|----------|---|
| A | Removing W from the circuit. |
| B | Adjusting W to a higher resistance. |
| C | Connecting a resistor parallel to the galvanometer. |
| D | Replacing wire PQ with another wire of similar length and resistivity but smaller diameter. |

Answer: B

Regardless, the p.d. across PJ (V_{PJ}) remains unchanged and equal to the e.m.f. of cell C.

Option A: Removing W increases the p.d. across PQ, hence increases the p.d. per unit length of PQ. Thus for the same V_{PJ} , balance length will decrease.

Option B: Adjusting W to a higher resistance decreases the p.d. across PQ, hence decreases the p.d. per unit length of PQ. Thus for the same V_{PJ} , balance length will increase.

Option C: No effect on the balance length.

Option D: If the wire has a smaller cross-sectional area, its resistance increases. With W unchanged in resistance, the p.d. across PQ increases, hence increases the p.d. per unit length of PQ. Thus for the same V_{PJ} , balance length will decrease.

24	X and Y are two coaxial flat circular coils lying on a table. Coil X has 120 turns and a diameter
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